Soft Value-Based Decoding for Guided Molecular Generation on a Frozen Graph-Diffusion Model

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1 Question

Reward-guided generation usually fine-tunes the generator (e.g., RL fine-tuning [2]), which is costly. Soft Value-Based Decoding (SVDD) [5] instead steers a frozen diffusion model at inference time: at every reverse step it draws M candidates, scores each with a value function, and keeps one. Its molecule experiments use a continuous model (GDSS) and defer DiGress to future work [5, App. C]. Whether SVDD works on a *discrete graph* diffusion model, and whether it beats drawing the same number of candidates as independent samples and keeping the best, is therefore open. **This project asks whether SVDD with a learned value function, applied to a frozen DiGress model trained on QM9, raises the mean synthetic-accessibility and QED reward of generated molecules more than best-of-N sampling with $N{=}M$ independent samples per slot, without lowering validity, and how that advantage scales with the number of candidates M.**

2 Primary Paper and Background

Primary: SVDD [5] shows that a soft value function $v_t(x_t) = \alpha \log \mathbb{E}[e^{r(x_0)/\alpha} \mid x_t]$ lets a pretrained diffusion model sample from a reward-tilted distribution with no fine-tuning, on images, DNA/RNA and molecules (default $M{=}10$, argmax selection). **Supporting:** DiGress [7] is a discrete graph diffusion model whose denoiser predicts clean atom and bond types (QM9: 99.0% valid, 96.2% unique, about 1 h to reach 99% validity). DDPO [2] is the fine-tuning route that SVDD avoids. Gao et al. [4] study best-of-n as a simple, strong reward-optimization baseline. TreeDiff [8] benchmarks SVDD and best-of-N on a different graph model (GraphDiT) under matched compute. **Gap:** the SVDD paper's molecule results use GDSS on ZINC-250k and defer DiGress to future work. I found no study of SVDD on DiGress/QM9 that counts network passes, compares against best-of-N over several seeds, and diagnoses the learned value function. I measure that. This is new work started this semester, not prior research.

3 Method

Pipeline (all inference-time; DiGress weights frozen). (1) *Proposal:* the official QM9 checkpoint link is dead (404; repository issue #109), so I train DiGress once on QM9 with the repository's unmodified QM9 config (cross-entropy on node/edge types, $T{=}500$) and then freeze it. A thin wrapper exposes $p_\theta(G_{t-1} \mid G_t) = \sum_{G_0} q(G_{t-1} \mid G_t, G_0)\, p_\theta(G_0 \mid G_t)$ and returns M i.i.d. candidates per molecule per step; one network pass gives $p_\theta(G_0 \mid G_t)$ for all M draws. (2) *Value function* $\hat{v}_\phi(G_t, t)$, trained by MSE regression onto $r(G_0)$ using 10k unguided rollouts (the SVDD-MC estimator); two architectures: an MLP on pooled atom/bond-type histograms plus a timestep embedding, and a 3-layer message-passing GNN on the (soft) graph. (3) *Selection:* keep $\arg\max_m \hat{v}_\phi(G_{t-1}^{(m)})$ (or sample $\propto e^{\hat{v}/\alpha}$). A training-free variant, SVDD-PM, scores a candidate by $r(\hat{G}_0)$ of its decoded denoiser prediction (one extra pass per candidate). (4) *Rewards:* SA reward $(10 - \mathrm{SA})/9$ from the SA score [3] and QED [1] via RDKit, invalid $\to 0$. Value functions are trained per reward. **Stretch:** a docking-proxy reward, dropped if not started by Nov 20 (see Risks).

4 Data

QM9 [6]: 133,885 molecules (up to 9 heavy atoms; the MoleculeNet CSV that DiGress downloads), public, CC0 on figshare, no personal data. DiGress trains on its own seeded 100k/val/test split of QM9 (repository code); afterwards I only *generate*. All of QM9 is the novelty reference. Three seeded (42), disjoint 1k index splits (committed) only sanity-check reward code and value-function predictions at $t{=}0$ on real molecules, never training or tuning. Value-function data are unguided rollouts from the frozen DiGress (separate training/validation chains and random streams from evaluation). Each evaluation run generates $N{=}2{,}000$ molecules.

5 Evaluation Plan

Metrics: mean and top-10% reward, validity (RDKit-sanitizable and connected; stricter than DiGress's largest-fragment convention, which I also report), uniqueness, novelty vs. QM9, diversity (mean pairwise ECFP4 Tanimoto distance), network passes and wall-clock per molecule. **Seeds:** 0, 1, 2 for every run, plus 3, 4 for the headline point ($M{=}10$); report mean±std. **Baselines:** (B1) unguided DiGress ($M{=}1$); (B2) *best-of-N*, $N{=}M$: draw M unguided molecules per slot and keep the highest-reward one (true reward, a strong baseline). It matches SVDD in candidates but spends up to $M\times$ more network passes, so I also report reward against network passes and wall-clock, including best-of-N at SVDD's wall-clock; (B3) SVDD-PM without a learned value. **Controlled comparisons:** (A1) $M \in \{1, 5, 10, 20\}$ for SA and QED; (A2) value estimator $\in$ {PM, MLP, GNN} at $M{=}10$. Not run: DDPO-style fine-tuning (outside the budget) and DiGress's regressor guidance (needs a retrained regressor; a different question). **Falsifiable outcome:** the hypothesis is wrong if, at $M{=}10$, SVDD-MLP's mean-reward gain over B2 is not larger than twice the across-seed standard deviation (the larger of the two methods'), *or* if SVDD reaches its gain only with validity more than 5 points below B1. A null or negative result will be reported and analyzed.

6 Compute Plan

Where: GAIVI (class partition, Slurm; scripts in repo), Colab/local as backup; I already have a GAIVI account; first job in week 1. **Estimate** per run = $Np/512 \times 0.1\,\text{s} \times 1.5/3600$ h, with N=2000 molecules, p network passes per molecule (T=500 for unguided and SVDD-MLP, $(M{+}1)T$ for SVDD-PM, MT for best-of-M), batches of 512 graphs, an assumed 0.1 s per batch-pass (≤9-node graphs; *measured in week 1*) and 1.5× overhead for featurizing and scoring; one $p{=}T$ run costs 0.08 h. B1 (5 seeds): 0.4 h. B2: one pool of 20 draws per slot per seed yields best-of-5/10/20 for both rewards: $3 \times 1.6 + 2 \times 0.8 = 6.5$ h. A1 (24 runs ×0.08 h, plus per (reward, seed) 11k rollout chains 0.45 h and one fit 0.5 h, assumed): 7.6 h. A2: SVDD-PM at M=10 and a random-selection control 5.9 h; GNN arm 6.2 h. Seeds 3–4 for SVDD-MLP at M=10: 4.1 h. Subtotal 30.7 h; with 25% margin 38 h; plus DiGress training on QM9, budget ≤10 GPU-h (paper: about 1 h to 99% validity): **≈48 GPU-hours**. **Single-GPU fallback (<2 h):** N=1000, $M \in \{1, 10\}$, one seed, MLP value, SA reward, on a short-trained DiGress checkpoint: two runs 0.04 h each, 1.2k rollout chains 0.05 h, one fit 0.5 h. Training checkpoints are kept and sampling is saved after every chunk of 512 molecules (a preempted run resumes to identical molecules); every run logs config, seed, git commit, GPU and metrics.

7 Milestones (approval Oct 12 → presentation, targeting Slot A, Fri Nov 13)

Oct 12–18	GAIVI environment (Python 3.9, torch 2.0.1); pin commits; first GPU job; **launch DiGress QM9 training**; measure s/epoch, s/batch-pass. Sign up for Slot A on Oct 13.
Oct 19–25	Finish training and reproduce DiGress validity (≥97%, paper 99.0%); finish the `propose()` wrapper with exactness tests; B1 over 3 seeds. **Gate Oct 25.**
Oct 26–30	Collect rollouts; train MLP value fn and check it at t=0; SVDD-PM end to end; B2; complete the <2 h fallback run. **Gate Oct 30:** go/no-go for Slot A (else ask for Slot C).
Oct 31–Nov 8	**Main comparison at** M=10, SA and QED, 3 seeds: SVDD-MLP vs. B2, B1, SVDD-PM; start the M sweep (A1). Optional Nov 6 office-hours check-in. Results frozen Nov 8.
Nov 9–13	Slides from frozen results (due Nov 12), rehearse code walk-through; **present Nov 13** (pipeline, main result, baselines; ablations shown as planned/partial).
Nov 14–22	Finish A1 (all M, both rewards); A2 incl. GNN value fn (**go/no-go Nov 20**); seeds 3–4 at M=10.
Nov 23–29	Failure analysis (where the value fn mis-ranks), figures, seed variance checks; results frozen Nov 29.
Nov 30–Dec 11	Report and clean-clone check; **report and code due Dec 11**.

8 Risk and Fallback

(i) If my DiGress QM9 model is below 97% validity (DiGress's metric) by Oct 25, I train longer or try the repository's `fixed_bug` branch (total extra budget ≤10 GPU-h); if it stays below 90%, I freeze it as is and report B1's lower validity (the 5-point rule is relative to B1). (ii) If the GNN value function does not beat the MLP on validation MSE by Nov 20, I drop it and keep PM vs. MLP as ablation A2; the docking proxy is dropped if not started by Nov 20. (iii) If measured s/batch-pass is more than 3× my estimate, I drop seeds 3–4, then cut N to 1000 and the M grid to $\{1, 10, 20\}$ so the total stays near 50 GPU-h. (iv) If SVDD does not beat B2, the deliverable becomes a negative-result analysis of the value function's ranking error. (v) If the Oct 30 gate fails, I ask to present in Slot C (Dec 4), same order of work.

9 Contribution Table (full version in `CONTRIBUTIONS.md`)

Component	Origin	What I changed
DiGress model	`cvignac/DiGress` @ `bdfa10c` (MIT)	Code and default QM9 config unmodified (no official checkpoint available); I train it once, then freeze; added a candidate-proposal wrapper (`vgmg/models/digress.py`)
SVDD algorithm	Li et al. [5]; `hyanan16/SVDD-molecule` @ `455e4d4` (GDSS/ZINC-250k, no license) read for reference only	Own implementation from the paper (`vgmg/rl/svdd.py`); ported to discrete graph diffusion
Value functions	Mine	MLP, GNN, and PM estimators; training loop
Rewards, metrics	RDKit QED; SA score from RDKit Contrib	Own wrappers and rescaling; validity/novelty/diversity code is mine
Best-of-N, ablation harness	Mine	Compute-matched baseline; sweep scripts and logging

10 AI Tool Declaration

I used Claude Code to scaffold the repository, draft unit tests, and edit this proposal. The research question, experimental design, baselines, and ablation choices are mine; I will write and understand the DiGress wrapper and SVDD code, and all reported numbers will come from my own runs.

References

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[6] Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific Data*, 1:140022, 2014.

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